ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–4: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation, The Atomic Foundations of Matter, Journey Inside the Atom** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
3	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
4	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
5	**(a)** 3 x 10^5	2 x 10^30 / 6 x 10^24.
6	**(a)** 50 per cent	30/60.
7	**(d)** 2 : 1	75/25 = 3; 60/40 = 1.5; ratio 2 : 1.
8	**(a)** S^{2-}	Z = 16 is sulphur, 2 extra electrons.
9	**(b)** the theory is strongly supported but remains open to tests	Science is provisional.
10	**(a)** only one component sublimes	Different behaviour on heating.
11	**(a)** 4.4 g	Mass of gas escaped.
12	**(c)** 24.3 u	(1896 + 250 + 286)/100 = 24.32.
13	**(d)** A is false but R is true.	A is false (theories can be refined); R is true.
14	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
15	**(c)** A is true but R is false.	A is true; R is false because the gas escapes.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Area = 96 square cm (1); About 2 per cent (1).	1+1
17	Mass of solution = 240 g (1); 24/240 x 100 = 10 per cent (1).	1+1
18	4 g H needs 32 g O (1); Water = 36 g; 8 g of oxygen left (1).	1+1
19	63x + 65(100 - x) = 6350 (1); x = 75, so the heavier isotope is 25 per cent (1).	1+1
20	A straight line shows proportionality (1); Easy to read the slope and spot errors (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Mean = 42.6/3 (1); = 14.2 cm (1); Greatest deviation = 0.1 cm (1).	1+1+1
22	320 g contains 120 g salt and 200 g water (1); At 20 degrees Celsius 200 g water holds 50 g (1); Crystals = 120 - 50 = 70 g (1).	1+1+1
23	S: 1/32 = 0.03125 (1); O: 1.6/16 = 0.1 (1); Ratio about 1 : 3: SO_3 (1).	1+1+1
24	2, 8, 3 (1); M^{3+} (1); Mass = 54 + 48 = 102 u (1).	1+1+1
	OR: Shows how reliable the result is (1); e.g. 12.3 +/- 0.1 cm (1); Allows comparison with other results (1).	1+1+1
25	A: 5/55 = 9.1 per cent (1); B: 8/108 = 7.4 per cent (1); A is more concentrated (1).	1+1+1
	OR: Mg2+ and O2- attract more strongly (1); More energy to break the lattice (1); Hence higher melting point (about 2800 vs 801 degrees Celsius) (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

26	Apparatus: thread, bob, stand, stopwatch, scale (1); Independent: length; dependent: time period (1); Constants: mass, small angle (1); Time 20 oscillations and repeat (1); Plot T against L and conclude that T increases with L (1).	1+1+1+1+1
	OR: NaCl 36 g stays dissolved at 20 degrees Celsius (solubility 36) (1); KNO3 remaining dissolved = 32 g (1); KNO3 crystallised = 110 - 32 = 78 g (1); Crystals are nearly pure KNO3 (1); Fractional crystallisation (1).	1+1+1+1+1
27	Run 1: O = 1.6 g, Mg : O = 3 : 2 (1); Run 2: O = 4.0 g, Mg : O = 3 : 2 (1); Same ratio in both runs (1); Formula MgO: 24 : 16 = 3 : 2 (1); Law and formula verified (1).	1+1+1+1+1
	OR: 69x + 71(100 - x) = 6980 (1); x = 60, so 60 per cent and 40 per cent (1); Neutrons: 38 and 40 (1); Element is gallium (1); Average mass between 69 and 71 (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–4: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation, The Atomic Foundations of Matter, Journey Inside the Atom** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(a)** 5.4 cm	Subtract the zero error.
3	**(b)** 32 g	Water 100 g holds 32 g; crystals 64 - 32.
4	**(b)** 3 : 8, CO_2	Atoms ratio 2.4/12 : 6.4/16 = 1 : 2.
5	**(b)** 20 and 16	X lost 2, Y gained 2.
6	**(a)** dosage of a medicine for a child	Safety critical.
7	**(a)** iron and sulphur	Only iron is magnetic.
8	**(b)** 78.5	52.5/47.5 x 71 = 78.5.
9	**(a)** 20	10x + 11(100 - x) = 1080.
10	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
11	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
12	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
13	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
14	**(d)** 1 per cent	0.2/20.4.
15	**(b)** 18 g	50 - 32.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Boiling point is fixed (1); Heat supplies latent heat (1).	1+1
17	Na+: 2, 8; F-: 2, 8 (1); Same as neon (1).	1+1
18	Maximum 2n squared with n = 1 is 2 (1); K shell is the smallest and nearest to the nucleus (1).	1+1
19	Distance = 3 x 10^5 x 3.15 x 10^7 (1); = 9.45 x 10^12 km (1).	1+1
20	Slow cooling allows ordered growth (1); Few nuclei form (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	p + n = 31 and n = p + 5 (1); 2p = 26, so p = 13 and n = 18 (1); Z = 13; configuration 2, 8, 3 (1).	1+1+1
22	Rising curve (1); Not a straight line (1); Longer pendulum has longer period (1).	1+1+1
23	Salt gives ions that neutralise the charge on particles (1); Particles aggregate (1); Soap separates (1).	1+1+1
24	Empirical mass = 30 (1); 60/30 = 2 (1); Molecular formula $C_2H_4O_2$ (1).	1+1+1
	OR: Electrons occupy fixed shells (1); No radiation in these states (1); Energy changes only by jumps (1).	1+1+1
25	Mean = 54.4 s for 30 (1); T = 1.81 s (1); Spread about 0.4 / 54.4 = 0.7 per cent (1).	1+1+1
	OR: X = 0.27 (1); Y = 0.6, Z = 0.8 (1); X is most strongly adsorbed (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Same mass and volume (1); Vary only temperature (1); Record times (1); Graph of time vs temperature (1); Errors: reaction time and stirring (1).	1+1+1+1+1

	OR: Particles carry charge and repel (1); Coagulation with electrolytes (1); Electrophoresis moves particles in a field (1); Dialysis separates dissolved ions (1); Examples: blood purification (1).	1+1+1+1+1
27	Atoms of different elements have different masses (1); 12 g of Mg has half as many atoms as 12 g of C (1); Number of atoms is proportional to mass divided by atomic mass (1); 24 g of Mg has as many atoms as 12 g of C (1); Atomic masses link mass to number of atoms (1).	1+1+1+1+1
	OR: n = 24 - 12 = 12 (1); B: p = 11, n = 12, A = 23 (1); B is sodium $^{23}_{11}\mathrm{Na}$ (1); 2, 8, 1 (1); A is magnesium-24 (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–4: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation, The Atomic Foundations of Matter, Journey Inside the Atom** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** N_2O	63.6/14 : 36.4/16 = 4.54 : 2.27 = 2 : 1.
2	**(d)** argon	18 electrons.
3	**(a)** check the experiment and the assumptions in the model	Investigate.
4	**(b)** dissolve in water, filter, evaporate	Sugar dissolves.
5	**(c)** 1 : 2 : 1	CH2O.
6	**(a)** 1/8	Three half-lives.
7	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
8	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
9	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
10	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
11	**(c)** 15 per cent	3 x 5 per cent.
12	**(b)** the one with R_f 0.9	Largest R_f.
13	**(c)** 310 u	120 + 62 + 128.
14	**(a)** 5	Z = 15: 2, 8, 5.
15	**(c)** to test the procedure and find problems	Trial run.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	117/58.5 = 2 (1); Twice as many (1).	1+1
17	No dense positive centre in his model (1); It predicted only small deflections (1).	1+1
18	Mean = 9.76 m/s^2 (1); Range = 0.3 m/s^2 (1).	1+1
19	Solute may have low solubility (1); e.g. a saturated solution of chalk or CO2 in water is dilute (1).	1+1
20	Cu: 80/64 = 1.25; O: 20/16 = 1.25 (1); Ratio 1 : 1: CuO (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Mass of O2 = 24.5 - 14.9 = 9.6 g (1); Percentage = 9.6/24.5 x 100 (1); = 39.2 per cent (1).	1+1+1
22	Same valence electrons in a group (1); Across a period valence electrons increase 1 to 8 (1); Chemical behaviour depends on valence electrons (1).	1+1+1
23	Volume = 144 cubic m (1); Mass = 144 x 1.2 (1); = about 173 kg (1).	1+1+1
24	Ink spots from document and suspects (1); Run chromatography (1); Compare bands and R_f values (1).	1+1+1
	OR: Weighing error (1); Leakage of a gas (1); The law holds when all products are weighed (1).	1+1+1
25	Splits on neutron capture giving energy (1); Chain reaction controlled in reactors (1); Radioactive waste and accidents (1).	1+1+1
	OR: Informed consent (1); Safety tests first (1); Honest reporting of side effects (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

26	Strength: evidence-based (1); Strength: self-correcting (1); Limitation: cannot test untestable ideas (1); Chance: penicillin discovery (1); Persistence: many experiments repeated (1).	1+1+1+1+1
	OR: Spot unknown and known dyes on the same paper (1); Run in a suitable solvent (1); Compare R_f values (1); Same R_f means same dye (1); Solvent changes R_f (1).	1+1+1+1+1
27	Conservation: atoms rearranged (1); Constant proportions: fixed ratios of atoms (1); Atoms are divisible (1); Isotopes differ in mass (1); Atoms combine in whole-number ratios remains valid (1).	1+1+1+1+1
	OR: Closest approach of alpha particles gives an upper limit for the nucleus size (1); Atom radius about 10^-10 m (1); Density = mass/volume is huge (1); About 10^18 kg per cubic metre (1); Nucleus holds nearly all the mass (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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